

Name: _____

Date: _____

FRACTIONS, DECIMALS AND PERCENTAGES, (PERCENTAGES AS FRACTIONS AND DECIMALS)

LO: I can convert between percentages, fractions and decimals fluently.

Three forms of the same value

Every **percentage** can be written as a **fraction** (over 100, then simplify) or a **decimal** (divide by 100).

Tip: Learn key equivalents: $\frac{1}{2} = 50\% = 0.5$, $\frac{1}{4} = 25\% = 0.25$, $\frac{1}{5} = 20\% = 0.2$, $\frac{1}{10} = 10\% = 0.1$.

Key equivalents

	$50\% = \frac{1}{2} = 0.5$	half
	$20\% = \frac{1}{5} = 0.2$	fifth
	$125\% = \frac{5}{4} = 1.25$	more than 100%
	$33\% = \frac{1}{3}$	$\frac{1}{3} = 33.3\ldots\%$
	$0.5\% = \frac{1}{200}$	$0.5\% = \frac{1}{200}$

Key idea

% to fraction: put over 100, simplify. % to decimal: divide by 100. Reverse to go back.

$$\textcircled{45\%} = \frac{45}{100} = \frac{9}{20} = \textcircled{0.45}$$

Percentage means per hundred

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - percentage to fraction

Convert 60% to a fraction in simplest form.

Write as $\frac{60}{100}$

HCF of 60 and 100 is 20

$$\frac{60}{100} = \frac{3}{5}$$

$$= \frac{3}{5}$$

Divide numerator and denominator by the HCF.

Example 2 - percentage to decimal

Convert 37.5% to a decimal.

Divide by 100

$$37.5 \div 100 = 0.375$$

$$= 0.375$$

$$= 0.375$$

Move the decimal point two places left.

Example 3 - fraction to percentage

Convert $\frac{7}{20}$ to a percentage.

Divide: $7 \div 20 = 0.35$

Multiply by 100: $0.35 \times 100 = 35$

$$= 35\%$$

$$= 35\%$$

Or multiply top and bottom to get denominator 100.

Example 4 - decimal to percentage

Convert 0.065 to a percentage.

Multiply by 100

$$0.065 \times 100 = 6.5$$

$$= 6.5\%$$

$$= 6.5\%$$

Move decimal point two places right.

YOUR TURN – PRACTICE (deliberate practice)

1. Percentages to fractions

Convert to a fraction in simplest form.

- a) 25%
- b) 40%
- c) 80%
- d) 15%

2. Percentages to decimals

Convert each percentage to a decimal.

- a) 70%
- b) 4%
- c) 12.5%
- d) 150%

3. Fractions to percentages

Convert to a percentage.

- a) $\frac{3}{4}$
- b) $\frac{2}{5}$
- c) $\frac{7}{10}$
- d) $\frac{5}{8}$

4. Decimals to percentages

Convert each decimal to a percentage.

- a) 0.35
- b) 0.08
- c) 1.2
- d) 0.005

5. Mixed conversions

Convert to the form shown in brackets.

a) 45% (fraction)

b) $\frac{3}{8}$ (decimal)

c) 0.72 (%)

d) 175% (fraction)

e) $1\frac{1}{20}$ (%)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $50\% = 0.5 = \frac{1}{2}$ ☐

Correct answer: _____

Reason: _____

b) $0.7 = 7\%$ ☐

Correct answer: _____

Reason: _____

c) $125\% = 1.25$ ☐

Correct answer: _____

Reason: _____

d) $\frac{1}{3} = 33\%$ ☐

Correct answer: _____

Reason: _____

e) To convert a decimal to %, multiply by 100 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Arrange in order from smallest to largest: $\frac{5}{8}$, 0.6, 63%, $1\frac{1}{20}$. Show your conversion working.

Expression: _____

Simplified: _____

b) A student says ' $\frac{1}{3} = 33\%$ '. Explain why this is not exactly correct. What is $\frac{1}{3}$ as an exact percentage?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

